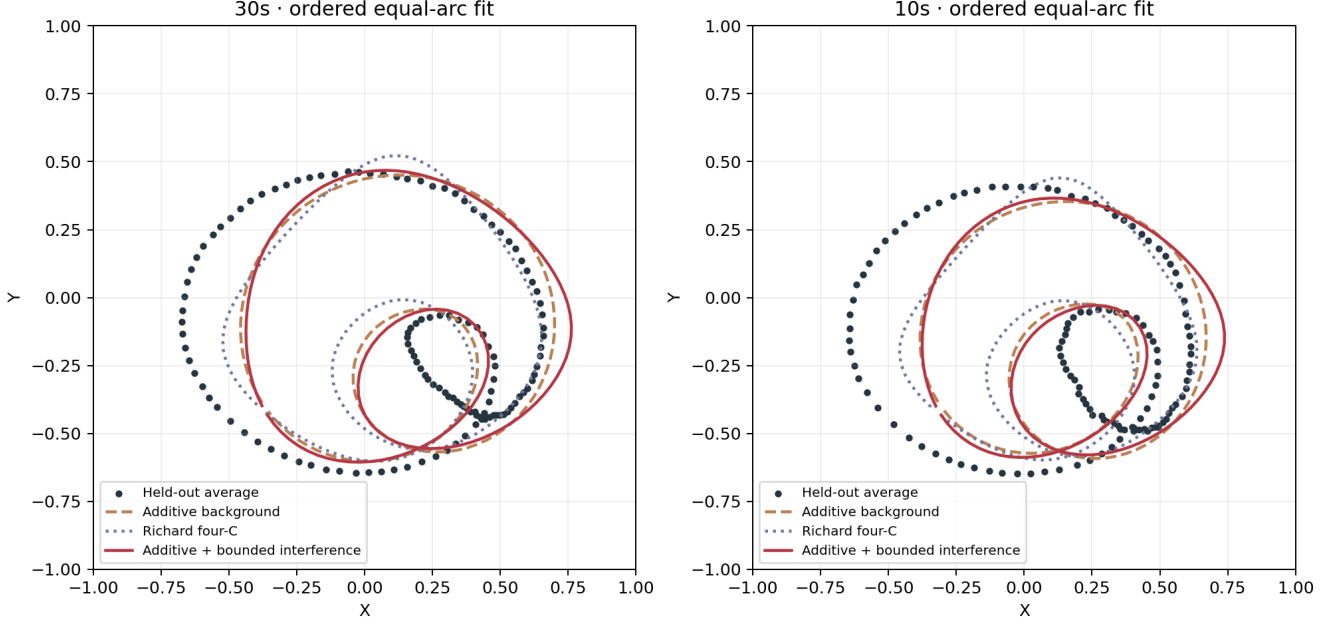


More complex background and inverse-circle tests

6 October 2026

Main findings. Adding bounded interference modestly improves the equal-arc intensity fit but leaves the inner-loop discrepancy. Direct signal inversion and circle fitting is computationally possible, and a known synthetic cone is recovered to numerical precision. Circle residual alone, however, admits a collapse/cancellation solution. A constrained inverse reconstruction remains an exploratory branch-specific test, not a validated replacement.



Window	Equal-arc model	Channel RMS	XY RMS	BG/ $T_{\max}$	θ range
30s	Additive	0.02990	0.14642	34.1%	37.8–90.0°
30s	Richard four-C	0.03641	0.16645	—	36.6–74.6°
30s	Additive + interference	0.02780	0.14591	32.4%	38.2–90.0°
10s	Additive	0.03146	0.15442	37.1%	37.6–90.0°
10s	Richard four-C	0.03827	0.17479	—	40.1–73.6°
10s	Additive + interference	0.02992	0.15446	34.8%	38.3–89.8°

The complex model is a separate channel-effective hypothesis, not Richard’s exact four-C model:

$$I_i = S_i + B_i + 2\rho_i\sqrt{B_iS_i}, \quad |\rho_i| \leq 1.$$

It fits three additive parameters (equal pair sums, independent asymmetries) and four constant signed overlaps. The bound prevents a cross term larger than the scalar coherent-field limit; it does not prove realization by a single common optical background field. Background cap remains 95% of minimum training total. Cone geometry, one brightness and one cyclic offset add five parameters: 12 in total. No flexible phase mapping is fitted. Thirty-two starts across both directions; selected arc fits converge. Calibration, NA .38–1.3, fixed excitation, 128 calibrated bins and relative channel-intensity objective are unchanged.

Inversion and the best circle without explicit cone parameters

For a candidate background, let $C_i = 2\rho_i\sqrt{B_i}$ (or Richard's independently fitted coefficient). On the unique positive-root domain $I_i - B_i > 0$,

$$S_i = \left[\frac{\sqrt{C_i^2 + 4(I_i - B_i)} - C_i}{2} \right]^2.$$

Compute corrected X/Y , then use

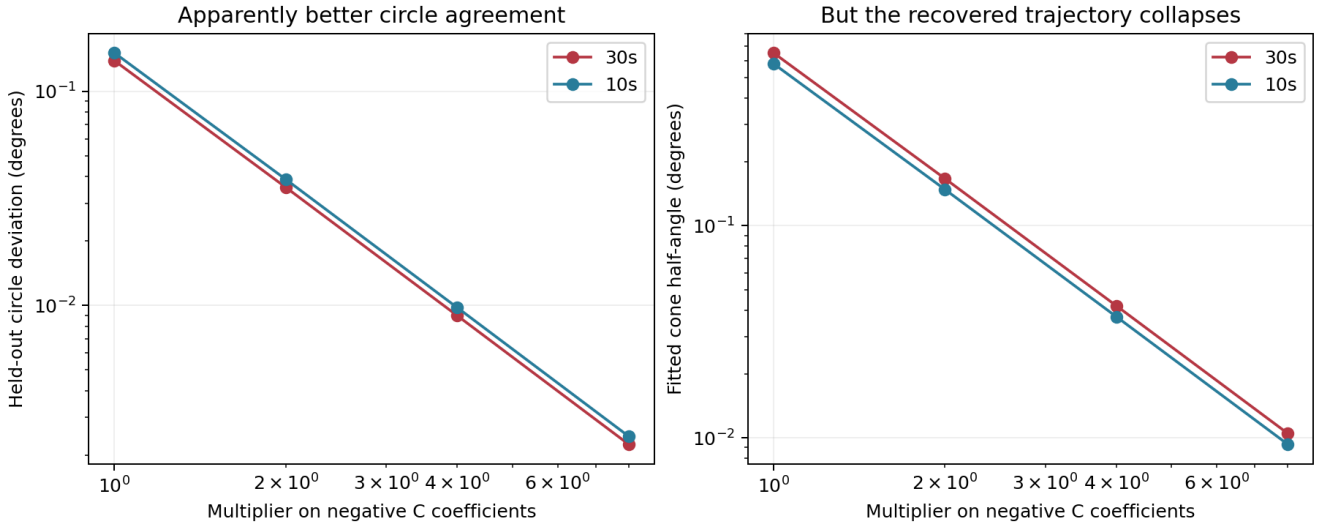
$$\sin^2 \theta = \frac{A_F r}{C_F - B_F r}, \quad \phi = \frac{1}{2} \text{unwrap}[\text{atan2}(Y, X)].$$

A small circle has plane equation $\mathbf{n} \cdot \mathbf{d} = c$, with $\|\mathbf{n}\| = 1$. For recovered directions $\mathbf{d}_k$, subtract their mean and form the covariance. Its smallest-eigenvalue eigenvector gives the least-squares plane normal; $c = \mathbf{n} \cdot \bar{\mathbf{d}}$. This minimizes squared plane distance. It is an algebraic circle fit, not an exact angular-distance minimizer. Angular deviations are reported separately.

Thus the cone is still implicit in the circle: axis $\mathbf{n}$, half-angle $\arccos c$. These quantities are recalculated from each candidate correction, not independent optimizer parameters. No per-point phase variables are optimized and no point order is changed.

Synthetic verification. A simulated cone with half-angle 0.22 radians, fixed brightness and known additive/interference correction was inverted. Maximum signal error 1.11×10^{-16} , maximum plane error 3.33×10^{-16} , reconstructed channel RMS 2.70×10^{-16} . This verifies the implementation on a valid non-crossing branch, not identifiability on real data.

Observed failure of circle-only fitting. Richard's free signed coefficients run toward large negative values, producing a large recovered signal almost cancelled by interference. The recovered direction changes very little. At 30–31 s, a fitted example has a cone half-angle of only 0.652° , corrected theta 36.33 – 37.88 degrees, and angular circle RMS 0.139° . Multiplying all four negative coefficients by eight, without further optimization, reduces half-angle to 0.0105° , theta span to about 0.025° , and circle error to 0.00224° . This is not evidence of accurate angles.



The corrected pair sums disagree by roughly 11% of corrected total at 30 s (15% at 10 s), despite the near-perfect circle. A common-brightness optical signal cannot explain those recovered four channels. This illustrates why circle agreement must retain the information discarded by the two anisotropy ratios. The initial numerical coefficient box was $[-5, 5]$; the scaling test shows that this box merely truncates the collapse, not that it resolves it.

Retaining intensity information in the inverse approach

A second pilot profiles the plane, projects each recovered direction onto its circle, evaluates the full four-channel model with one fitted brightness scale, and minimizes the original relative-intensity residual. Phase locations are inferred from each observed four-channel vector, not independently adjusted parameters.

Window	Inverse reconstruction	Channel RMS	BG/ $T_{\max}$	Recovered θ	Circle RMS
30s	Additive	0.06448	16.3%	5.8–84.7°	7.13°
30s	Richard four-C	0.05982	—	8.2–76.8°	7.25°
30s	Additive + interference	0.06448	16.3%	5.8–84.7°	7.13°
10s	Additive	0.06160	17.9%	8.1–80.8°	7.21°
10s	Richard four-C	0.05849	—	7.6–71.9°	6.64°
10s	Additive + interference	0.04839	20.1%	10.5–79.5°	8.80°

Interpretation limits. These are conditional held-out reconstruction scores: background, plane and brightness are trained on training cycles, but each held-out vector supplies its own recovered direction and hence phase location. They are not the same prediction test as evaluating a fixed curve at predetermined held-out bin indices. The 30 s complex result retains the nested additive solution because the searches did not improve its feasible training cost; it is not a separately converged complex optimum.

The strict pilot requires nonnegative corrected roots, anisotropy below the Fourkas maximum, and a closed continuous lift in one hemisphere. It retains acquisition order; no nearest-point permutation is allowed. The inferred phase is checked for backtracking; selected reconstruction examples show none or only small fluctuations. Hemisphere-crossing branches are not globally searched. These searches do not prove that a more complete inverse treatment cannot work.

The first unconstrained trials included nonclosing lifted trajectories and out-of-domain points. They are retained as diagnostic failures, not accepted angle corrections. Some circle-only complex solutions were feasible on training data but exceeded the physical anisotropy range on held-out data; these cannot be interpreted by clipping theta. Strict reconstructed examples above have valid held-out inversions.

Why branch handling matters. Fourkas inversion determines folded theta and phi modulo pi. A collection of individually admissible points is not automatically a continuous closed physical trajectory. A cone crossing the equator also folds when represented only on the upper hemisphere. Eliminating explicit cone parameters does not eliminate these geometric decisions.

Recommended interpretation. Keep the equal-arc fits as the controlled comparison. The bounded interference extension tested here does not resolve the shape mismatch. The inverse route is useful as a diagnostic, but a raw spherical-circle objective is demonstrably insufficient: it must enforce four-channel optical consistency, brightness and branch continuity, or explicitly incorporate their measurement uncertainty. Neither route establishes absolute angle truth from these data.

Scripts, caches and numerical details are in `model_comparison/inverse_circle/`. No commit or push.